

PRICING ANALYST PORTFOLIO PROJECT

Solar PPA Pricing & Discounting

What a public market benchmark can and can't tell you about whether a deal was priced well

THE SITUATION

A pricing team needs to know if a deal was priced well — and what to trust when checking

There's no public dataset of real negotiated deals with win/loss outcomes, so this project uses a strong analog: LBNL's utility-scale solar PPA database, benchmarked against a third-party market-price index — the same kind of check a pricing desk runs before signing off on a deal.

THE QUESTION

Are recent solar PPA deals priced well relative to the market — and can a public price index actually answer that?

Data & approach

LBNL project data

1,775 real utility-scale solar projects, 2007–2024. Genuinely structured — a clean Excel workbook, not scraped prose.

LevelTen & Trio index

Third-party regional PPA price benchmark, 7 organized markets. Confirmed its actual methodology before trusting it — see the diagnosis ahead.

Method

Filter to disclosed-price PV projects, join to the regional benchmark, compare — then check whether that comparison is actually fair.

1,759 PV projects after excluding CSP. Only 425 (24%) disclose a PPA price at all — a real selection-bias constraint kept in view throughout.

FINDING 1 — AN 85% PRICE DECLINE, VERIFIED

~85%

drop in median disclosed PPA price, 2010–11 vintage (\$~200/MWh) to 2022–24 vintage (~\$25–32/MWh)

Tracks median LCOE closely over the same period — independently reproducing LBNL's own published industry trend, a useful sanity check before trusting anything more specific.

FINDING 2 — A MISTAKE CAUGHT BEFORE IT SHIPPED

The first version of this analysis said 2011 projects were overpriced by **+\$194/MWh**. That number is wrong.

It came from comparing 2011-vintage project prices directly against the 2024 market benchmark — two prices from cost eras 14 years and an 85% industry decline apart. Not a pricing signal. A vintage mismatch.

Wrong approach

Compare ALL vintages (2007–2024) against the single 2024 benchmark

Fixed approach

Restrict the benchmark comparison to matching-vintage (2022–24) projects only

FINDING 3 — EVEN FAIRLY COMPARED, EVERY SEGMENT IS BELOW BENCHMARK

107 vintage-matched (2022–24) projects, compared against the 2024 index

Segment	n	vs. benchmark
PJM (region)	15	−\$28.59/MWh
Corporate (offtaker)	8	−\$33.35/MWh
Public Utilities (offtaker)	19	−\$22.72/MWh
CAISO (region)	21	−\$18.19/MWh
Monopoly IOU (offtaker)	5	−\$16.46/MWh
MISO (region)	13	−\$13.80/MWh
Investor-owned Utility (offtaker)	11	−\$11.00/MWh

Every reliably-sized segment ($n \geq 5$, benchmark available) — not one scattered around zero. That consistency is the actual signal.

THE DIAGNOSIS — WHY

An asking price and a closing price are not the same measurement.

*“The offers underlying our Index are from projects that are currently under development and posted by developers to the LevelTen Marketplace... giving the industry a transparent and unprecedented look at actual PPA price offers — **not estimates of what PPA prices could or should be.**”*

— LevelTen Energy, PPA Price Index methodology

The index is a **P25 offer price** for projects still seeking a buyer. LBNL's data is a **realized, signed price** for completed projects. A closing price coming in below an asking-price index is close to the expected outcome — not evidence of negotiation skill.

RECOMMENDATION

Check what a benchmark measures before trusting a comparison built on it.

1. Match the deal stage

Don't compare a signed price against an offer/asking-price index without adjustment. Use a signed-deal comparator, or an internal same-vintage benchmark, if one is genuinely needed.

2. Always vintage-match

This project's own first pass got this wrong (+\$194/MWh) before it was caught. Check vintage alignment before any two-price comparison, every time.

3. State the disclosure rate

Only 24% of PV projects disclose a PPA price. Say so up front in any summary — don't let a reader assume full market coverage.

LIMITATIONS & ASSUMPTIONS

- CSP plants excluded (16 of 1,775 projects) — PPA structure isn't directly comparable to PV
- 184 of 425 disclosed-price projects (43%) sit in regions with **no matching third-party benchmark at all** (non-ISO West, non-ISO Southeast, Hawaii)
- **NYISO's own benchmark value is an Excel formula error** in LBNL's source workbook — genuinely unavailable, not a small-sample issue
- Segment sample sizes remain modest even after the vintage fix (n=5 to n=21) — real numbers, not large enough for strong causal claims about *why* any one segment differs
- This is a public-data analog for a deal-pricing problem — there's no "lost deal" in this dataset, only completed projects

WHAT I'D DO WITH MORE DATA OR TIME

This pre-empts the obvious follow-up questions

- Get a signed-deal comparator (a paid LevelTen tier, if one includes realized prices) to actually test the diagnosis rather than infer it from published methodology alone
- Investigate why disclosure itself is selective — is the 24% that discloses systematically different from the 76% that doesn't?
- Extend the offtaker-type and region segmentation once more vintage-matched data accumulates past 2024
- Build a small interactive dashboard so a reader can filter by region/vintage themselves, rather than only seeing the segments chosen here

APPENDIX — DATA & SOURCES

Source	What	Freq.
LBNL Utility-Scale Solar, 2025 Edition	Project-level PPA price, CapEx, LCOE, capacity factor	Annual, 1,775 projects
LevelTen & Trio 2024 PPA Index	Regional P25 offer-price benchmark, 7 ISO/RTO markets	Annual

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